Reviewer 1 Comments - Technical and Methodological Focus

Overall Assessment

This manuscript presents PharmacoProfiler, a web platform for pharmacogenomic data integration and drug response prediction. While the work addresses an important need in the field, significant technical and methodological concerns prevent acceptance in its current form.

Recommendation: Major Revision Required

Major Concerns

1. Insufficient Technical Documentation

The methodology section lacks critical details necessary for reproducibility:

- **Machine Learning Model**: No description of algorithm selection, hyperparameter tuning, or model architecture
- **Data Preprocessing**: Missing details on normalization methods, outlier handling, and quality control measures
- **Feature Engineering**: No explanation of how molecular features are extracted and selected
- **Cross-validation Strategy**: Unclear how model performance was assessed and validated

Recommendation: Provide comprehensive technical documentation including:

- Detailed ML pipeline description
- Feature selection methodology
- Hyperparameter optimization approach
- Cross-validation and external validation strategies

2. Limited Performance Evaluation

The reported Pearson correlations (0.6 intra-platform, 0.4 cross-platform) lack context:

- **No Baseline Comparisons**: How does this compare to existing methods?
- **Statistical Significance**: No confidence intervals or significance tests provided
- **Performance Metrics**: Only Pearson correlation reported; need RMSE, MAE, R^2
- **Validation Scope**: Limited evidence of robust external validation

Recommendation: Conduct comprehensive benchmarking including:

- Comparison with state-of-the-art methods
- Multiple performance metrics

- Statistical significance testing
- External validation on independent datasets

3. Data Integration Methodology

While data harmonization is claimed as a key contribution, the process is poorly described:

- **Identifier Mapping**: How are compounds and cell lines mapped across databases?
- **Data Standardization**: What normalization methods are applied?
- **Quality Control**: How are inconsistencies and outliers handled?
- **Batch Effects**: Are systematic differences between datasets addressed?

Recommendation: Provide detailed data integration pipeline with:

- Step-by-step harmonization process
- Quality control measures
- Batch effect correction methods
- Data validation procedures

4. Network Visualization Validation

The tripartite network approach lacks validation:

- **Network Properties**: No analysis of network topology or properties
- **Biological Relevance**: Limited evidence that network connections are meaningful
- **Visualization Effectiveness**: No user studies or usability testing
- **Scalability**: How does visualization perform with large networks?

Recommendation: Validate network approach through:

- Network topology analysis
- Biological pathway enrichment
- User experience studies
- Scalability testing

Minor Issues

5. Writing Quality and Presentation

- **Incomplete Sections**: Several sections appear truncated or incomplete
- **Author Affiliations**: Missing complete institutional information

- **Figure Quality**: Figures need better resolution and detailed captions
- **Reference Formatting**: Inconsistent citation format throughout

6. Reproducibility Concerns

- **Code Availability**: No mention of code or data availability
- **Implementation Details**: Insufficient information to reproduce results
- **Platform Access**: Limited information about platform accessibility

Specific Technical Questions

1. **Algorithm Selection**: Why was the chosen ML algorithm selected over alternatives?
2. **Feature Importance**: Which molecular features are most predictive?
3. **Model Interpretability**: How can users understand prediction rationale?
4. **Computational Requirements**: What are the hardware/software requirements?
5. **Update Frequency**: How often is the underlying data updated?

Suggested Improvements

Immediate Actions Required

1. **Expand Methodology**: Add comprehensive technical section (2-3 pages)
2. **Benchmarking Study**: Compare against existing platforms and methods
3. **Validation Experiments**: Conduct external validation on independent datasets
4. **User Studies**: Evaluate platform usability and effectiveness

Medium-term Enhancements

1. **Model Interpretability**: Add SHAP or LIME analysis for prediction explanation
2. **Uncertainty Quantification**: Provide confidence intervals for predictions
3. **Batch Processing**: Enable bulk prediction capabilities
4. **API Development**: Provide programmatic access to platform features

Minor Corrections Needed

Abstract

- Specify exact number of datasets integrated (currently says "six major sources")
- Clarify what "harmonizing" means in this context
- Add specific performance metrics beyond Pearson correlation

Introduction

- Provide clearer motivation for tripartite network approach
- Better positioning relative to existing platforms
- Quantify the "fragmentation" problem being addressed

Results Section (appears incomplete)

- Add comprehensive performance evaluation
- Include comparative analysis with existing tools
- Provide detailed case studies demonstrating platform utility

Conclusion

This work has potential but requires substantial revision to meet publication standards. The technical contributions are incremental rather than groundbreaking, but the platform could provide value to the research community if properly validated and documented.

Time Estimate for Revision: 4-6 months for comprehensive revision including additional experiments and validation studies.

Recommendation: Major revision with emphasis on technical rigor, comprehensive validation, and detailed methodology documentation.

Reviewer 2 Comments - Biological Relevance and Clinical Impact Focus

Overall Assessment

The authors present PharmacoProfiler, a web platform integrating pharmacogenomic data for drug response prediction and network visualization. While the platform addresses a relevant problem in cancer pharmacogenomics, several concerns regarding biological interpretation, clinical relevance, and comparative positioning limit its current impact.

Recommendation: Major Revision Required

Major Strengths

1. Comprehensive Data Integration

The integration of multiple major pharmacogenomic databases (GDSC, CCLE, NCI-60, gCSI, FIMM) represents a valuable resource consolidation effort that could benefit the research community.

2. Novel Visualization Approach

The tripartite network visualization connecting cell lines, drugs, and indications offers an intuitive way to explore complex pharmacological relationships, which is lacking in current platforms.

3. Practical Utility

The combination of data exploration and predictive modeling in a single platform addresses real researcher needs for drug repurposing and biomarker discovery.

Major Concerns

1. Limited Biological Validation

Critical Issue: The manuscript lacks sufficient biological validation of the platform's predictions and network relationships.

Specific Concerns:

- **Pathway Analysis**: No evidence that predicted drug-cell line associations align with known biological pathways
- **Mechanism Validation**: Limited discussion of how predictions relate to drug mechanisms of action
- **Biomarker Relevance**: No analysis of whether identified associations correspond to clinically relevant biomarkers
- **False Discovery Rate**: No assessment of prediction reliability or false positive rates

Recommendations:

- Conduct pathway enrichment analysis for predicted associations
- Validate predictions against known drug-target interactions
- Analyze correspondence with clinical biomarkers
- Implement statistical significance testing for network connections

2. Clinical Relevance Assessment

Major Gap: Insufficient demonstration of clinical applicability and translational potential.

Missing Elements:

- **Clinical Correlation**: No analysis of how cell line predictions relate to patient outcomes
- **Biomarker Translation**: Limited discussion of translating findings to clinical settings
- **Drug Development Impact**: Unclear how platform supports actual drug development decisions

- **Regulatory Considerations**: No discussion of regulatory implications for predictions

Recommendations:

- Include analysis correlating cell line sensitivity with patient response data
- Discuss limitations of cell line models for clinical prediction
- Provide case studies demonstrating clinical relevance
- Address regulatory and ethical considerations

3. Comparative Analysis Deficiency

Significant Weakness: Inadequate comparison with existing platforms and methods.

Missing Comparisons:

- **Platform Functionality**: No systematic comparison with PharmacoDB, CellMinerCDB, or DepMap
- **Prediction Performance**: No benchmarking against other drug response prediction methods
- **Unique Value Proposition**: Unclear advantages over existing solutions
- **User Experience**: No comparison of usability or adoption metrics

Recommendations:

- Conduct head-to-head comparison with major existing platforms
- Benchmark prediction performance against published methods
- Clearly articulate unique advantages and limitations
- Include user feedback and adoption metrics

4. Data Quality and Standardization Issues

Important Concern: Limited discussion of data quality control and standardization challenges.

Specific Issues:

- **Batch Effects**: No mention of addressing systematic differences between datasets
- **Data Quality Metrics**: Missing information on data filtering and quality control
- **Standardization Methods**: Unclear how different assay types are normalized
- **Missing Data Handling**: No discussion of how incomplete data is managed

Recommendations:

- Implement and document comprehensive quality control procedures

- Address batch effect correction explicitly
- Provide data quality metrics and filtering criteria
- Discuss limitations imposed by data heterogeneity

Moderate Concerns

5. Network Analysis Depth

The tripartite network approach, while novel, lacks rigorous analysis:

- **Network Properties**: No analysis of network topology, clustering, or centrality measures
- **Biological Interpretation**: Limited discussion of what network patterns mean biologically
- **Statistical Validation**: No statistical testing of network structure significance
- **Dynamic Analysis**: No consideration of how networks change with different filters

6. Machine Learning Model Limitations

- **Model Interpretability**: Limited discussion of how users can understand predictions
- **Uncertainty Quantification**: No confidence intervals or prediction reliability measures
- **Feature Importance**: Missing analysis of which molecular features drive predictions
- **Model Generalizability**: Unclear how well models transfer across cancer types

Minor Issues

7. Presentation and Documentation

- **Figure Quality**: Network visualizations need better resolution and clearer legends
- **Case Studies**: Missing detailed examples demonstrating platform capabilities
- **User Documentation**: Insufficient information about platform usage and interpretation
- **Accessibility**: Limited discussion of platform availability and sustainability

8. Technical Implementation

- **Scalability**: No discussion of computational requirements or performance limits
- **Update Mechanisms**: Unclear how data is maintained and updated

- **Quality Assurance**: Missing information about ongoing validation and maintenance

Specific Biological Questions

1. **Cancer Type Specificity**: How do predictions vary across different cancer types, and is this biologically meaningful?
2. **Drug Mechanism Correlation**: Do predicted sensitivities align with known drug mechanisms and target expressions?
3. **Resistance Mechanisms**: Can the platform identify or predict drug resistance mechanisms?
4. **Combination Therapy**: Does the platform support analysis of drug combinations or polypharmacology?
5. **Biomarker Discovery**: What novel biomarkers have been identified through the platform, and how have they been validated?

Suggested Improvements

High Priority

1. **Biological Validation Study**: Conduct comprehensive validation of predictions against known biology
2. **Clinical Correlation Analysis**: Analyze relationship between cell line predictions and patient outcomes
3. **Comparative Benchmarking**: Systematic comparison with existing platforms and methods
4. **Quality Control Documentation**: Detailed description of data quality measures

Medium Priority

1. **Case Study Development**: Detailed examples demonstrating biological insights
2. **User Experience Study**: Evaluation of platform usability and researcher adoption
3. **Network Analysis Enhancement**: Rigorous analysis of network properties and significance
4. **Model Interpretability**: Implementation of prediction explanation methods

Recommendations for Revision

Content Additions Needed

1. **Biological Validation Section**: 2-3 pages demonstrating biological relevance

2. **Clinical Relevance Discussion**: Analysis of translational potential and limitations
3. **Comparative Analysis**: Systematic comparison with existing solutions
4. **Quality Control Methods**: Detailed data processing and validation procedures

Experimental Validation Required

1. **Pathway Enrichment Analysis**: Validate predictions against biological pathways
2. **Clinical Correlation Study**: Compare predictions with patient response data
3. **Benchmarking Experiments**: Performance comparison with existing methods
4. **User Studies**: Platform usability and effectiveness evaluation

Conclusion

This work addresses an important need in pharmacogenomics but requires substantial revision to demonstrate biological relevance and clinical impact. The platform shows promise but needs rigorous validation and better positioning relative to existing solutions.

Estimated Revision Time: 4-6 months including additional validation experiments and comparative studies.

Final Recommendation: Major revision with emphasis on biological validation, clinical relevance demonstration, and comprehensive comparative analysis.

Reviewer 3 Comments - Platform Design, Usability, and Presentation Focus

Overall Assessment

The manuscript describes PharmacoProfiler, a web-based platform for pharmacogenomic data integration and analysis. While the platform concept is sound and addresses a genuine need, significant issues with presentation quality, platform documentation, and user experience design limit its potential impact and adoption.

Recommendation: Major Revision Required

Major Strengths

1. User-Centered Design Philosophy

The integration of interactive visualization, filtering capabilities, and predictive modeling in a single web platform demonstrates good understanding of researcher workflow needs.

2. Comprehensive Data Scope

The platform's coverage of 50,000 compounds across 2,000 cell lines from multiple major databases represents impressive data consolidation that could save researchers significant time.

3. Practical Features

The combination of network exploration, filtering options, high-resolution exports, and prediction capabilities addresses multiple stages of the drug discovery research pipeline.

Major Concerns

1. Inadequate Platform Documentation

Critical Issue: The manuscript provides insufficient information about platform design, architecture, and user experience.

Missing Documentation:

- **User Interface Design**: No screenshots or detailed description of the interface
- **Workflow Examples**: Missing step-by-step usage scenarios
- **Feature Documentation**: Insufficient detail about filtering options and capabilities
- **Performance Specifications**: No information about response times, data limits, or system requirements
- **Accessibility Features**: No discussion of accessibility compliance or mobile responsiveness

Recommendations:

- Include comprehensive screenshots showing key platform features
- Provide detailed user workflow examples with step-by-step instructions
- Document all filtering and analysis options available to users
- Specify technical requirements and performance characteristics
- Address accessibility and cross-platform compatibility

2. Poor Manuscript Presentation Quality

Significant Issue: Multiple presentation problems undermine the manuscript's credibility and readability.

Specific Problems:

- **Incomplete Sections**: Several sections appear truncated or unfinished
- **Author Information**: Missing complete institutional affiliations for some authors
- **Figure Quality**: Figures are referenced but not properly presented or captioned
- **Reference Formatting**: Inconsistent citation style and incomplete references
- **Writing Quality**: Multiple grammatical errors and unclear sentences

Recommendations:

- Complete all truncated sections with full content
- Provide complete author information and affiliations
- Include high-quality figures with detailed captions
- Standardize reference formatting throughout
- Conduct thorough proofreading and editing

3. Insufficient User Experience Evaluation

Major Gap: No evidence of user testing, feedback collection, or usability evaluation.

Missing Elements:

- **User Studies**: No formal or informal user testing reported
- **Feedback Collection**: No evidence of researcher feedback incorporation
- **Usability Metrics**: No data on user task completion rates, error rates, or satisfaction
- **Learning Curve**: No assessment of how easily new users can adopt the platform
- **User Demographics**: No information about target user groups or requirements

Recommendations:

- Conduct formal usability testing with target researchers
- Collect and analyze user feedback systematically
- Report usability metrics and user satisfaction data
- Assess platform learning curve and onboarding effectiveness
- Define clear user personas and use cases

4. Limited Platform Sustainability Discussion

Important Concern: No discussion of long-term platform maintenance, updates, or sustainability.

Missing Information:

- **Maintenance Plan**: How will the platform be maintained and updated?
- **Data Currency**: How frequently will underlying data be refreshed?
- **Technical Support**: What support will be available to users?
- **Funding Model**: How will ongoing development and hosting be funded?
- **Community Engagement**: Plans for user community building and feedback

Recommendations:

- Develop and document platform sustainability plan
- Establish clear data update schedules and procedures
- Define user support mechanisms and resources
- Address long-term funding and maintenance strategies
- Create community engagement and feedback systems

Moderate Concerns

5. Platform Architecture and Scalability

Technical Concerns: Limited information about platform architecture and scalability.

Issues:

- **Technology Stack**: No description of underlying technologies used
- **Scalability Limits**: Unknown how platform performs with large datasets or many users
- **Data Storage**: No information about data storage and management approaches
- **Security**: Missing discussion of data security and privacy measures
- **Performance Optimization**: No evidence of performance testing or optimization

6. Integration and Interoperability

Usability Concerns: Limited discussion of how platform integrates with researcher workflows.

Missing Features:

- **Data Export**: Limited information about export formats and capabilities
- **API Access**: No mention of programmatic access options
- **Integration**: How does platform connect with other research tools?
- **Batch Processing**: Can users process multiple queries efficiently?
- **Collaboration**: Are there features supporting research team collaboration?

Minor Issues

7. Visual Design and Information Architecture

- **Network Visualization**: Limited discussion of visualization design principles
- **Information Hierarchy**: Unclear how information is organized and prioritized

- **Visual Consistency**: No evidence of systematic visual design approach
- **Color Accessibility**: No consideration of color-blind accessibility

8. Help and Documentation Systems

- **User Guidance**: No mention of help systems or user documentation
- **Tutorial Content**: Missing information about user onboarding and training
- **Error Handling**: No discussion of how user errors are handled and communicated
- **FAQ and Support**: No evidence of comprehensive user support resources

Specific Platform Questions

1. **User Onboarding**: How do new users learn to use the platform effectively?
2. **Data Interpretation**: What guidance is provided for interpreting results and predictions?
3. **Quality Control**: How are users informed about data quality and limitations?
4. **Performance Monitoring**: How is platform performance monitored and optimized?
5. **User Feedback**: What mechanisms exist for collecting and incorporating user feedback?

Suggested Improvements

High Priority Platform Enhancements

1. **User Interface Documentation**: Comprehensive screenshots and feature descriptions
2. **Usability Testing**: Formal evaluation with target user groups
3. **Performance Benchmarking**: Testing with realistic usage scenarios
4. **Help System Development**: Comprehensive user documentation and tutorials

Medium Priority Improvements

1. **API Development**: Programmatic access for power users
2. **Mobile Optimization**: Responsive design for tablet and mobile access
3. **Collaboration Features**: Tools for sharing and collaborating on analyses
4. **Advanced Visualization**: Enhanced network visualization options

Manuscript Revision Requirements

Content Additions Needed

1. **Platform Architecture Section**: Detailed technical description (1-2 pages)

2. **User Experience Evaluation**: Results from usability testing (1-2 pages)
3. **Interface Documentation**: Screenshots and feature descriptions (2-3 pages)
4. **Sustainability Plan**: Long-term maintenance and development strategy (1 page)

Presentation Improvements Required

1. **Complete All Sections**: Finish truncated content throughout manuscript
2. **Professional Figures**: High-quality screenshots and diagrams with detailed captions
3. **Consistent Formatting**: Standardize references, headings, and style throughout
4. **Thorough Editing**: Professional proofreading and editing for clarity and accuracy

Additional Materials Needed

1. **Supplementary Materials**: User manual, API documentation, technical specifications
2. **Video Demonstrations**: Screen recordings showing platform capabilities
3. **Case Study Details**: Extended examples with complete workflows
4. **User Testimonials**: Feedback from beta users or collaborators

Platform Access and Availability

Critical Issue: The manuscript does not provide clear information about platform access, availability, or terms of use.

Required Information:

- Platform URL and access requirements
- Registration process and user authentication
- Terms of service and data usage policies
- Availability for academic vs. commercial use
- System requirements for optimal performance

Conclusion

While PharmacoProfiler appears to be a potentially valuable platform for the pharmacogenomics community, the current manuscript presentation significantly undermines its impact. The platform concept is solid, but poor documentation, missing usability evaluation, and presentation quality issues prevent proper assessment of its contributions.

Key Message: This work needs substantial revision focusing on professional presentation, comprehensive platform documentation, and evidence of user-centered design validation.

Estimated Revision Time: 3-4 months focusing on presentation quality, platform documentation, and user experience evaluation.

Final Recommendation: Major revision with emphasis on professional presentation, comprehensive documentation, and user experience validation before resubmission.

Meta-Review and Editorial Decision - PharmacoProfiler Manuscript

Editorial Summary

After careful consideration of three independent reviewer reports, this meta-review provides the final editorial assessment of the manuscript "PharmacoProfiler: A web platform for harmonizing pharmacological data and AI-powered drug-response prediction" submitted for publication consideration.

Reviewer Consensus

All three reviewers acknowledge the potential value of the PharmacoProfiler platform while identifying significant concerns that prevent acceptance in its current form. The reviews demonstrate remarkable consistency in identifying core issues requiring attention.

Areas of Agreement

1. **Platform Concept**: All reviewers recognize the value of integrating multiple pharmacogenomic databases with interactive visualization
2. **Technical Gaps**: Universal concern about insufficient methodological documentation and validation
3. **Presentation Quality**: Consistent identification of poor manuscript presentation and incomplete sections
4. **Revision Scope**: Agreement that major revision (3-6 months) is required

Complementary Perspectives

- **Reviewer 1** (Technical Focus): Emphasized methodological rigor and reproducibility concerns
- **Reviewer 2** (Biological Focus): Highlighted biological validation and clinical relevance gaps
- **Reviewer 3** (Usability Focus): Identified platform documentation and user experience issues

Major Issues Requiring Resolution

1. Technical and Methodological Deficiencies

Critical Concerns:

- Incomplete machine learning methodology description
- Insufficient performance validation and benchmarking
- Missing comparative analysis with existing platforms
- Lack of reproducibility documentation

Required Actions:

- Comprehensive methodology section expansion (2-3 pages)
- Rigorous benchmarking against state-of-the-art methods
- External validation on independent datasets
- Code and data availability statements

2. Biological and Clinical Validation Gaps

Major Deficiencies:

- Limited biological interpretation of predictions
- No clinical correlation analysis
- Insufficient pathway validation
- Missing biomarker relevance assessment

Required Actions:

- Biological pathway enrichment analysis
- Clinical correlation studies where possible
- Validation against known drug-target interactions
- Discussion of clinical translation limitations

3. Presentation and Documentation Issues

Fundamental Problems:

- Incomplete manuscript sections
- Missing author affiliations
- Poor figure quality and documentation
- Inadequate platform usage documentation

Required Actions:

- Complete all truncated sections
- Professional manuscript editing and formatting

- High-quality figures with detailed captions
- Comprehensive platform documentation

Detailed Revision Requirements

Immediate Actions (Weeks 1-4)

1. **Complete Manuscript Structure**
    - Finish all truncated sections
    - Correct author affiliations
    - Standardize reference formatting
    - Professional editing and proofreading
2. **Methodology Enhancement**
    - Detailed ML pipeline description
    - Data preprocessing documentation
    - Cross-validation strategy explanation
    - Hyperparameter optimization details

Short-term Goals (Months 1-2)

3. **Performance Validation**
    - Comprehensive benchmarking study
    - Multiple performance metrics (RMSE, MAE, R^2)
    - Statistical significance testing
    - External validation experiments
4. **Platform Documentation**
    - User interface screenshots and descriptions
    - Workflow examples and tutorials
    - Technical specifications and requirements
    - Accessibility and compatibility information

Medium-term Objectives (Months 2-4)

5. **Biological Validation**
    - Pathway enrichment analysis
    - Drug-target interaction validation
    - Network topology analysis

    - Biological significance assessment
6. **Comparative Analysis**
    - Head-to-head comparison with existing platforms
    - Unique value proposition articulation
    - Advantages and limitations discussion
    - User experience comparison

Long-term Requirements (Months 4-6)

7. **User Experience Evaluation**
    - Formal usability testing
    - User feedback collection and analysis
    - Learning curve assessment
    - Adoption metrics reporting
8. **Sustainability Planning**
    - Long-term maintenance strategy
    - Data update procedures
    - Community engagement plan
    - Funding and support model

Specific Revision Guidance

Abstract Revision

Current Issues: Vague claims, incomplete metrics, unclear positioning **Required Changes**:

- Specific dataset numbers and validation metrics
- Clear statement of novel contributions
- Quantified performance improvements over existing methods

Introduction Enhancement

Current Issues: Poor positioning, incomplete background **Required Changes**:

- Comprehensive literature review
- Clear problem statement and motivation
- Detailed comparison with existing solutions
- Explicit statement of contributions and novelty

Methodology Section (New)

Required Content:

- Data collection and preprocessing pipeline
- Machine learning algorithm selection and justification
- Feature engineering and selection methods
- Cross-validation and evaluation strategies
- Statistical analysis procedures

Results Section (Major Expansion)

Required Content:

- Comprehensive performance evaluation
- Comparative analysis with existing methods
- Biological validation results
- Platform usage statistics and examples
- User feedback and adoption metrics

Discussion Enhancement

Required Content:

- Biological interpretation of findings
- Clinical translation potential and limitations
- Comparison with state-of-the-art approaches
- Future development roadmap
- Sustainability and maintenance plans

Quality Standards for Resubmission

Technical Rigor

- Reproducible methodology with complete documentation
- Rigorous statistical analysis and validation
- Comprehensive benchmarking against existing methods
- Clear evidence of biological and clinical relevance

Presentation Quality

- Professional manuscript formatting and editing
- High-quality figures with detailed captions

- Complete and accurate reference formatting
- Clear and engaging writing throughout

Platform Documentation

- Comprehensive user interface documentation
- Detailed usage examples and tutorials
- Technical specifications and requirements
- Evidence of user testing and feedback incorporation

Editorial Decision

Decision: MAJOR REVISION REQUIRED

Rationale

While the PharmacoProfiler platform addresses an important need in computational pharmacogenomics and demonstrates potential value for the research community, the current manuscript falls significantly short of publication standards for a high-impact journal. The combination of technical gaps, presentation issues, and insufficient validation prevents acceptance without substantial revision.

Revision Timeline

Recommended Timeline: 4-6 months for comprehensive revision **Minimum Acceptable Timeline**: 3 months with focused improvements **Maximum Allowable Timeline**: 8 months for complete restructuring

Resubmission Requirements

1. **Point-by-point response** to all reviewer comments
2. **Highlighted changes** in revised manuscript
3. **Supplementary materials** including user documentation and technical specifications
4. **Independent validation** results demonstrating platform effectiveness
5. **User study results** showing platform usability and adoption

Success Criteria for Resubmission

- Complete technical methodology documentation
- Rigorous performance validation and benchmarking
- Biological significance demonstration
- Professional presentation quality
- Comprehensive platform documentation

- Evidence of user testing and feedback incorporation

Final Recommendations

To Authors

1. **Prioritize Technical Rigor**: Focus first on completing methodology and validation sections
2. **Seek Collaboration**: Consider partnering with clinicians for biological validation
3. **Professional Editing**: Invest in professional manuscript editing services
4. **User Testing**: Conduct formal usability studies with target researchers
5. **Community Engagement**: Present platform at conferences to gather feedback

To Editorial Board

1. **Extended Review Timeline**: Allow 4-6 months for comprehensive revision
2. **Expert Consultation**: Consider additional expert reviewers for resubmission
3. **Platform Evaluation**: Request live platform demonstration for final review
4. **Impact Assessment**: Evaluate potential community impact and adoption likelihood

Conclusion

The PharmacoProfiler platform represents a valuable contribution to computational pharmacogenomics with significant potential for community impact. However, the current manuscript requires substantial revision across technical, biological, and presentation dimensions. With dedicated effort and comprehensive revision, this work could make an important contribution to the field.

Final Status: Major revision required with strong encouragement for resubmission following comprehensive improvements.

Editorial Confidence: High confidence in decision based on consistent reviewer feedback and clear revision pathway.

Expected Outcome: With proper revision, this manuscript has strong potential for acceptance and positive community impact.